

BIGBOY ADVANCE INSTALLATION INSTRUCTIONS

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1 Introduction

The Gameboy Macro is a popular mod, consisting of a Nintendo DS Lite console, with the top screen removed, that can be used to play Gameboy Advance games natively. For some, it is a cheap way to play their childhood favourite games on a better quality screen than what was offered on the original Gameboy Advance and Gameboy Advance SP.

In my opinion, the Gameboy Macro has three flaws. The first is that the GBA cartridge sticks out the bottom of the console, the second is that the screen is smaller than that of an original GBA and the third is the lack of “clicky” tactile buttons. With the BigBoy Advance mod, I aim to fix all three of these flaws.

The BigBoy Advance mod uses the Nintendo DS Lite motherboard to run Gameboy Advance games natively. The motherboard is housed in a custom designed shell that positions the GBA cartridge slot at the top of the console and does not cause the cartridge to stick out. The custom shell houses a custom button PCB that contains “clicky” GBA SP style buttons. The button presses are transferred to the motherboard via a ribbon cable connection. Additionally, the button PCB contains circuitry that allows a DSi XL bottom screen to be used with the DS Lite motherboard. With the larger screen, tactile buttons and hidden GBA cartridge, this mod fixes all three of the main flaws with a traditional Gameboy Macro mod.

2 Bill of materials

The BigBoy Advance kit can be purchased but does not include all components necessary to perform the mod. Some components are not included in the kit to give the modder creative freedom over the final aesthetic, for example, selecting the manufacturing method of the shell or choosing the colours of the buttons to be used. The only components that should be taken from the original Nintendo DS Lite console is the motherboard and cartridge shielding. The components included in the kit are given in Table 1. The components that are excluded from the kit are given in Table 2. These components should be sourced by the person performing the mod.

Table 1: Components included in BigBoy Advance kit

Component	Description	Quantity
BBA Button PCB	Button and screen adapter PCB that fits inside custom shell	1
BBA Button Extender Flex PCB	Flex PCB that connects the button PCB to the Nintendo DS Lite motherboard	1
Trigger Button PCB	PCB to mount trigger button inside custom shell.	2
Tactile switch	Tactile Switches 6 x 7.4 mm 5 mm Right Angle Through Hole SPST	2
FPC 26P 1.0mm	Clamshell FPC Connector, 26-pin, 1mm pitch	1
26 pin 1.0mm Ribbon Cable	26-pin, 1mm pitch, 10cm length, same-side	1
39pin 0.3mm ribbon cable	39-pin, 0.3mm pitch, 12cm length, same-side	1
SMD Resistor 330Ohm	330Ω 0805 Chip Resistor	1
Headphone Jack	SJ-3506-SMT-TR	1
Volume Wheel	B203, 20kΩ, 5 pin through hole	1
Speaker	2W, 8Ω, 23mm	1
Trigger dowels	∅2mm x 12mm shaft that acts as the axis of rotation for trigger buttons	2
Trigger button springs	Torsion springs used for GBA SP trigger buttons	2
Threaded inserts	M2 x 3.5mm OD x 3mm heat-set inserts	24
LED Light Pipes	PLP2-4MM Bivar Panel Mount 1-Way Straight Rigid LED Light Pipe Clear, 3 mm	2
Self-Tapping 2mm x 4mm	Phillips head pan head self-tapping screws. 4mm length.	2

Self-Tapping 2mm x 5mm	Phillips head pan head self-tapping screws. 5mm length.	23
M2 x 4mm Button Head Screw	ISO7380 M2 screws. 11x button PCB, 2x trigger PCB	13
M2 x 4mm Socket Head Screw	DIN912 M2 screws. 4x motherboard, 6x rear shell	10
M2 x 6mm Countersunk Screw	DIN7991 M2 screw. Used to secure battery cover.	1
Washer 2.2mm	Washer for M2 screws. Used to secure Nintendo DS Lite motherboard to rear shell.	4
M2 Nut	Secures battery cover when using a resin printed rear shell.	1

Table 2: Components excluded from BigBoy Advance kit

Component	Description	Quantity
Nintendo DS Lite Motherboard		1
Wire	Wire for connecting the trigger button PCB to the motherboard. Any wire gauge may be used.	
DSi XL Bottom Screen	Bottom LCD screen of a Nintendo DSi XL with touch digitizer removed	1
Screen lens	Screen lens of Nintendo DSi XL top screen	1
Front shell	3D printed part that forms part of the custom shell	1
Rear Shell	3D printed part that forms part of the custom shell	1
Battery Cover	3D printed part that forms part of the custom shell	1
Cartridge Shielding	Metal shielding removed from original Nintendo DS Lite console	1
GBA SP button set	Full button set compatible with GBA SP in preferred colour	1
Button Membrane set	Full set of button membranes for GBA SP in preferred colour	1

3 Tools Required

The following tools are required to assemble the BigBoy Advance:

- Soldering iron and solder
- Flux
- Wire strippers
- Phillips PH1 screwdriver
- 1.5 mm hex key
- 1.3 mm hex key
- Tweezers
- Flush cutters

4 Installation Instructions

4.1 Motherboard preparation

Before installation of the BigBoy Advance kit, the Nintendo DS Lite motherboard must be prepared by removing some components. The following components must be removed from the motherboard:

- Volume slider
- Power and charge indicator LEDs (3x LEDs in total)
- Charge port (OPTIONAL)

The components to be removed are shown, circled in red, in Figure 1 and Figure 2.

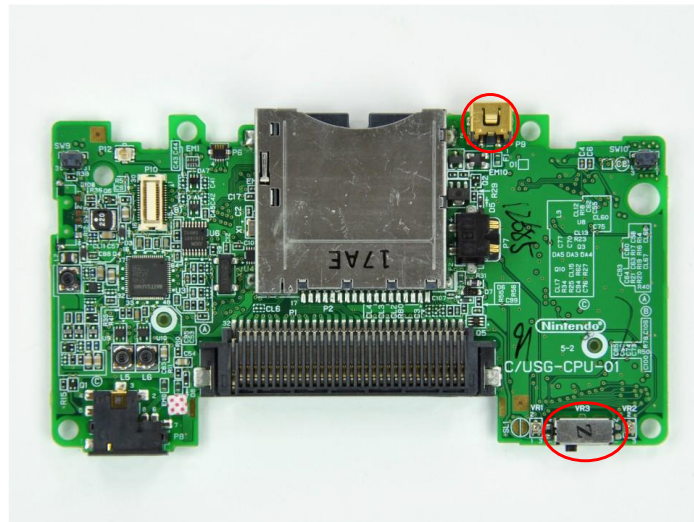


Figure 1: Components to remove from the back of the DS Lite motherboard

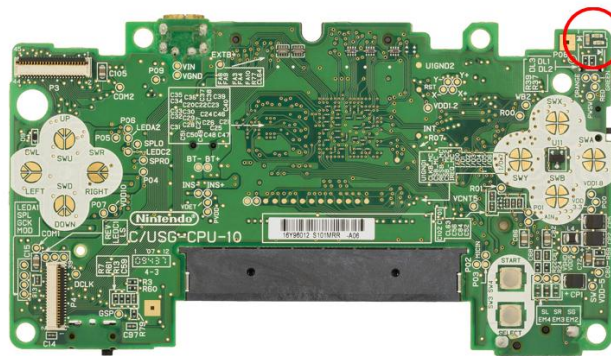


Figure 2: Components to remove from the front of the DS Lite motherboard

Figure 3 shows a motherboard with its volume slider and LEDs removed.

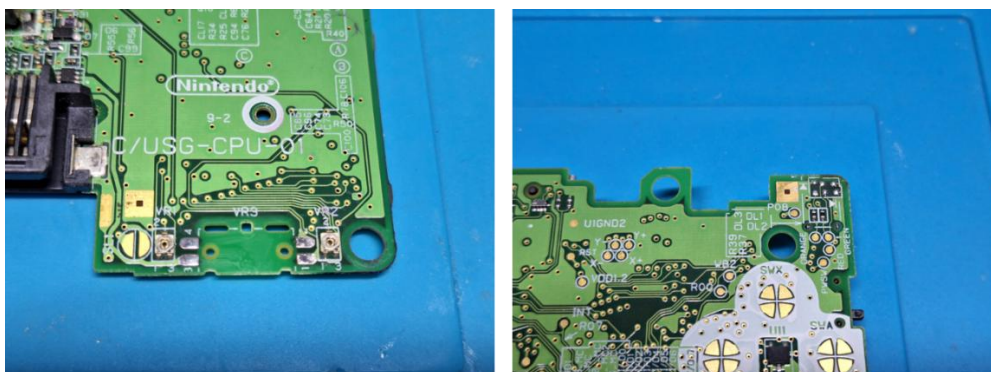


Figure 3: DS Lite motherboard after component removal

4.2 Shell preparation

The amount of preparation necessary for the shell parts is dependent on the manufacturing method used for the shells. Resin printed and aluminium shell parts don't require heat-set inserts to be installed. For resin printed shell parts, self-tapping screws are used to fasten components. For aluminium shell parts, the threads for machine screws are tapped during manufacturing.

NOTE: The shell design intended for resin printing can also be printed in nylon and assembled using self-tapping screws instead of heat-set inserts.

4.2.1 Rear Shell

1. **Resin (SLA):** Glue the battery retention M2 nut into the designated cavity in the shell as shown in Figure 4.

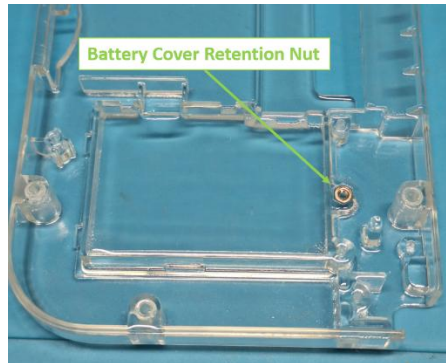


Figure 4: Resin printed shell battery compartment with retention nut

2. **Nylon (SLS/MJF):** Install seven heat-set threaded inserts using a soldering iron. Six of the inserts are installed on the inside of the shell as shown in Figure 5. The remaining insert is installed in the battery compartment as shown in Figure 6.

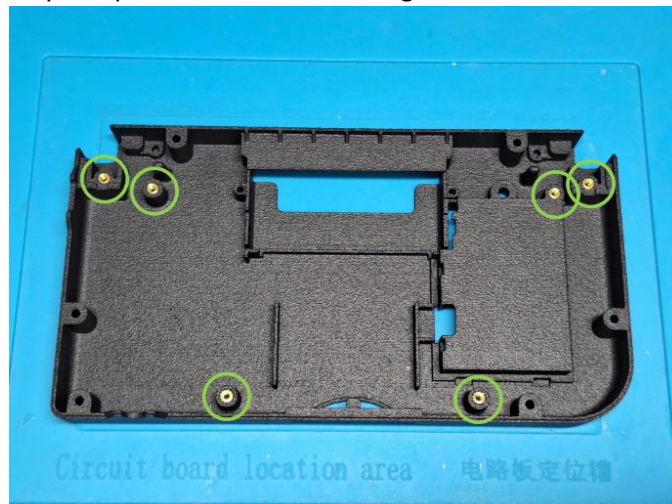


Figure 5: Heat-set inserts installed in rear shell

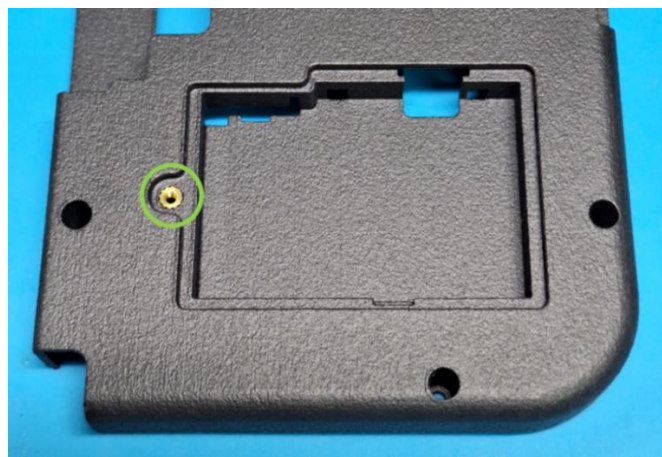


Figure 6: Heat-set insert installed in battery compartment

3. **Aluminium:** No additional thread preparation is required. Threaded holes for machine screws are provided during manufacturing.

4.2.2 Front Shell

1. Install two LED light pipes in the designated holes as shown in Figure 7. The light pipes are press fit.

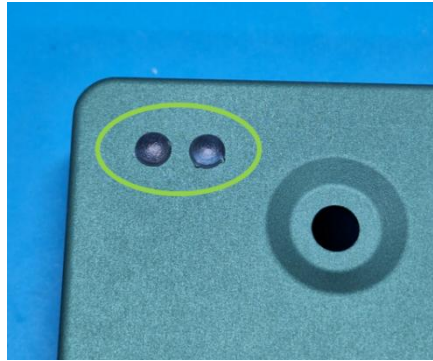


Figure 7: LED light pipes

2. **Nylon (SLS/MJF):** Install seventeen heat-set threaded inserts using a soldering iron as shown in Figure 8.

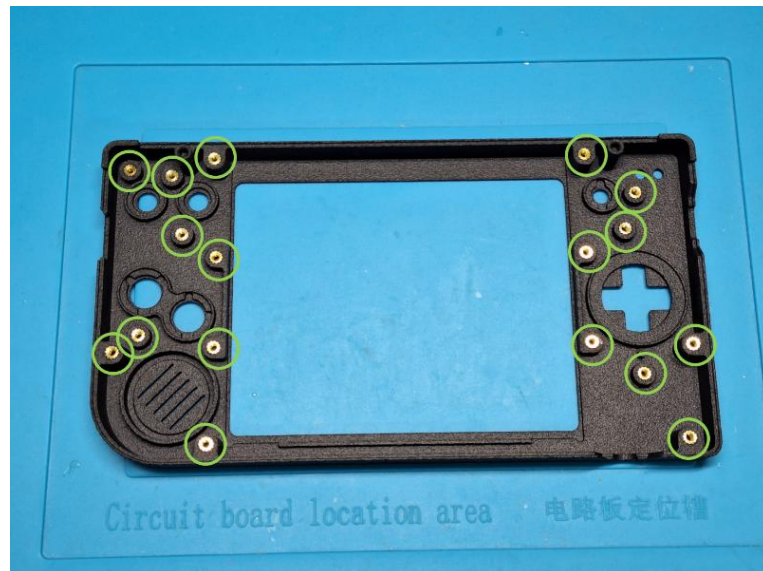


Figure 8: Heat-set inserts installed in front shell

3. **Resin (SLA):** No heat-set inserts are required. Components will be fastened using self-tapping screws.
4. **Aluminium:** No additional thread preparation is required. Threaded holes for machine screws are added during manufacturing.

4.3 Kit installation

The BigBoy Advance kit consists of a main button PCB, flex PCB and two trigger button PCBs. These components are fitted into a custom designed shell to create the ultimate native GBA playing experience. This section specifies the steps required to complete the mod successfully.

4.3.1 Front Shell Assembly

1. Solder the headphone jack and volume wheel to the button PCB as shown in Figure 9. Ensure that the components are soldered to the correct side of the PCB.

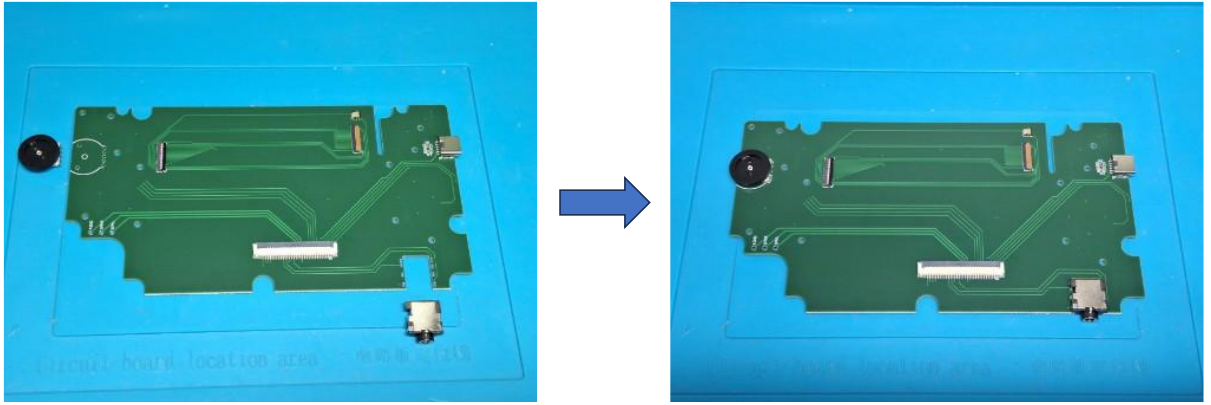


Figure 9: Volume wheel and headphone jack soldering

2. Place the screen into the front shell. The metal housing of the screen has a U-shape hook that fits over the screen retaining feature in the front shell as shown in Figure 10.

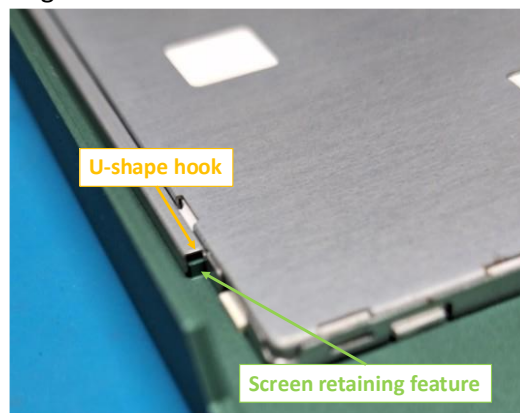


Figure 10: Screen retaining feature

3. Place the speaker into the speaker cavity as shown in Figure 11.



Figure 11: Speaker position in front shell

4. Place the buttons and membranes into the corresponding cavity in the front shell as shown in Figure 12.

NOTE: The brightness button of the GBA SP is used as the power button for the BigBoy Advance console.

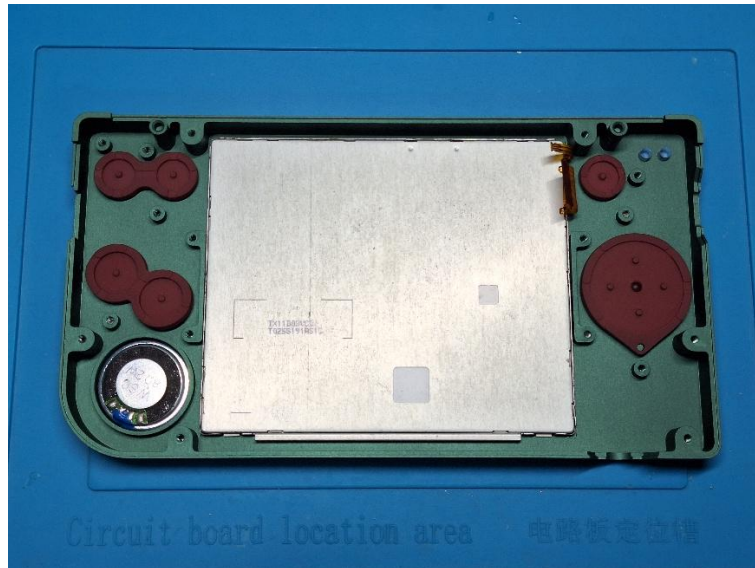


Figure 12: Buttons and membranes placed in front shell

5. Place the button PCB into the front shell. When placing the button PCB, ensure that the two ribbon cables of the screen feed through the designated cutout in the PCB without pinching as shown in Figure 13.

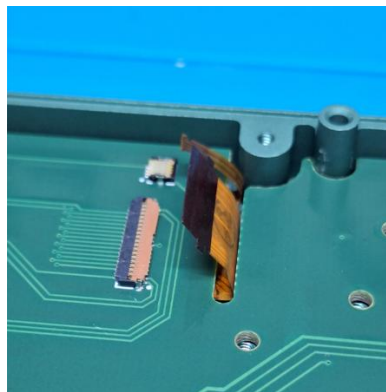


Figure 13: Cutout in button PCB for screen ribbon cables

6. Fix the button PCB to the front shell using eleven M2 x 4mm button head screws as shown in Figure 14.

NOTE: If using a resin printed shell, use 2mm x 5mm self-tapping screws instead.

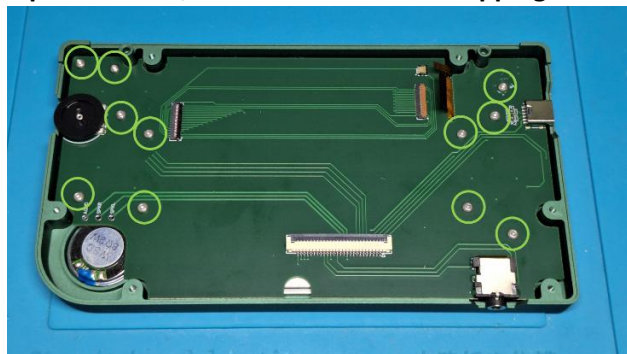


Figure 14: Button PCB fixed to front shell

7. Connect the two ribbon cables of the screen to the corresponding FPC connectors on the button PCB as shown in Figure 15.

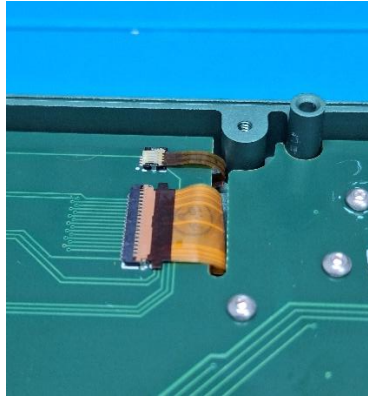


Figure 15: Screen ribbon cables connected to button PCB

8. Solder two short wires from the speaker to the button PCB as shown in Figure 16. The negative terminal of the speaker should be soldered to the AGND pad on the button PCB, and the positive terminal of the speaker should be soldered to the SPKL pad on the button PCB.

NOTE: In the next revision of the button PCB, I will combine the stereo audio channels into a mono audio channel.

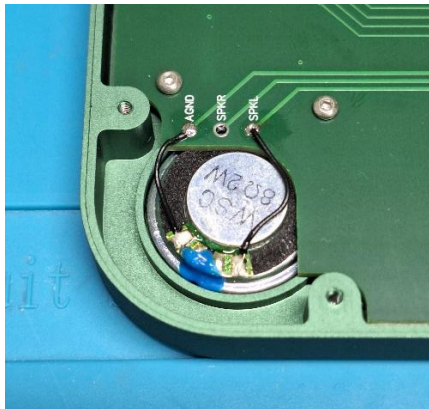


Figure 16: Speaker soldered to corresponding pads on button PCB

9. Install the screen lens for a Nintendo DSi XL top screen into the front shell as shown in Figure 17.

NOTE: The screen lens that is used is for the top screen of a Nintendo DSi XL.



Figure 17: Screen lens installed in front shell assembly

4.3.2 Rear Shell Assembly

1. Solder the 330 Ohm resistor and 26 pin FPC connector to the button extender flex PCB as shown in Figure 18.

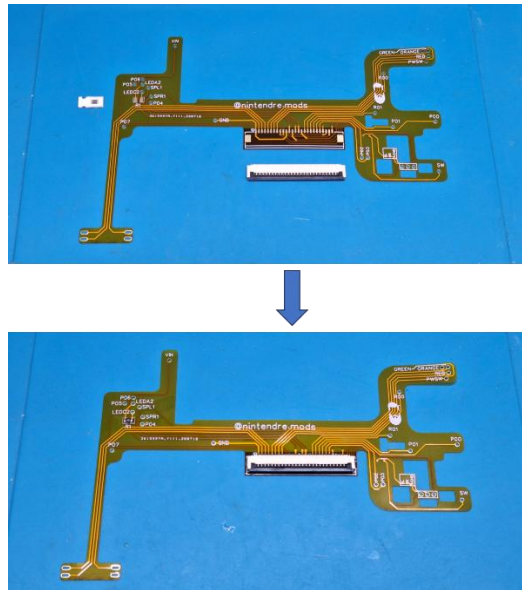


Figure 18: Button extender flex PCB with components

2. Solder the button extender flex PCB to the front of the DS Lite motherboard as shown in Figure 19. Ensure that the relevant through hole pads of the flex PCB line up with the corresponding test pads on the motherboard before soldering. The through hole pads on the flex PCB have identical designations to the test pads on the motherboard.

NOTE: If you did not remove the charging port from the motherboard, the pins of the charge port might prevent the flex PCB from sitting flush with the motherboard. You can trim away a small piece of the flex PCB to prevent this as shown in Figure 20.

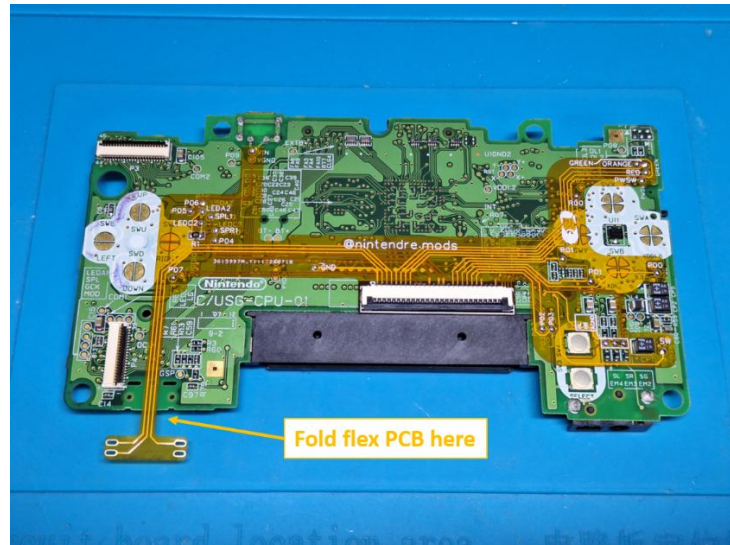


Figure 19: Button extender flex PCB soldered to DS Lite motherboard

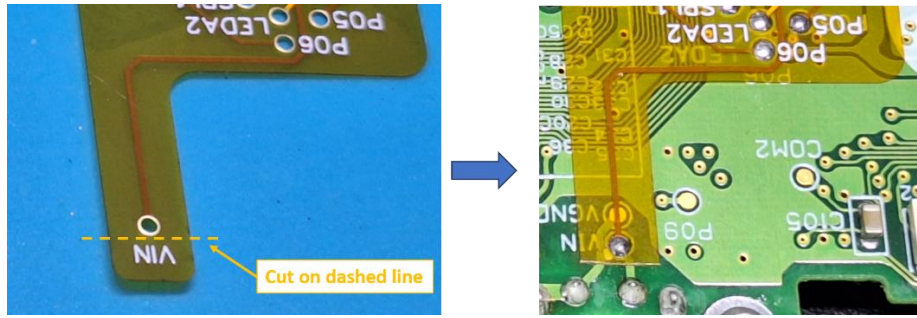


Figure 20: Cutting flex PCB when charge port was not removed

3. Fold the long leg of the button extender flex PCB over the bottom edge of the motherboard and solder the corresponding pads on the flex PCB to the pads of the volume slider on the motherboard as shown in Figure 21.



Figure 21: Button extender flex PCB soldered to volume slider pads

4. Solder a tactile button to each of the trigger button PCBs as shown in Figure 22. Ensure that the tactile button is soldered to the correct side of the PCB (opposite side than the silkscreen print).

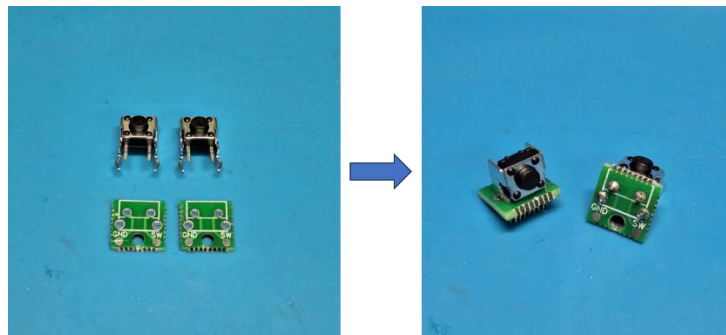


Figure 22: Trigger button PCBs with tactile buttons

5. Insert the two trigger dowels into the corresponding holes in the rear shell as shown in Figure 23.

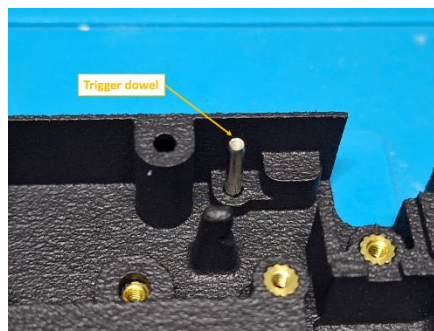


Figure 23: Trigger dowel installed in rear shell

6. Slide the right and left trigger buttons and trigger springs onto the corresponding trigger dowels. Hook the free leg of each trigger spring into the notched pillar to secure it as shown in Figure 24.

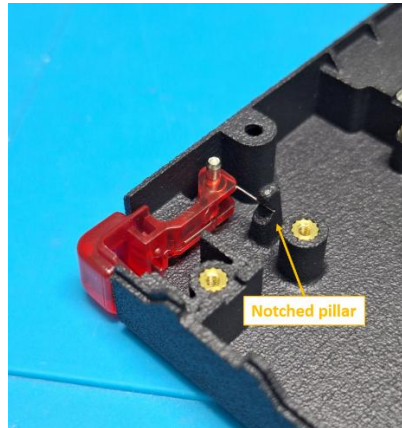


Figure 24: trigger button and spring installed in rear shell

7. Place the two trigger button PCBs into the rear shell and fasten each one with a M2 x 4mm button head screw as shown in Figure 25.

NOTE: If using a resin printed shell, use 2mm x 5mm self-tapping screws instead.



Figure 25: Trigger button PCB installed in rear shell

8. Place the cartridge shielding (taken from the Nintendo DS Lite console) and fasten it to the rear shell using two 2mm x 4mm self-tapping screws as shown in Figure 26.



Figure 26: Cartridge shielding installed in rear shell

9. Place the Nintendo DS Lite motherboard in position. The tabs of the Gameboy Advance cartridge slot and battery terminal should slide into the support geometry in the rear shell.

Fasten the motherboard to the rear shell using four M2 x 4mm socket cap head screws and four M2 washers.

NOTE 1: The screws can be button head or socket head cap screws. The kit includes socket head cap screws for this step. The screws used in Figure 27 are button head screws.

NOTE 2: If using a resin printed shell, use 2mm x 5mm self-tapping screws instead.

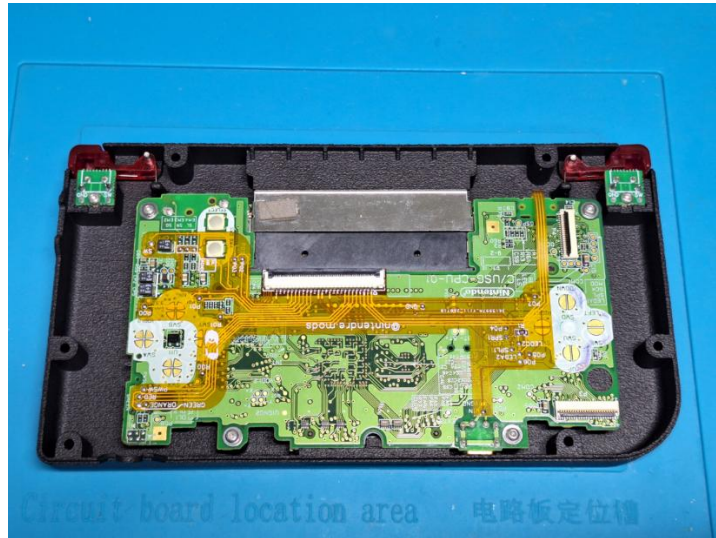


Figure 27: Nintendo DS Lite motherboard installed in rear shell

10. Solder a wire from the “SW” pad on the left trigger button PCB to the test pad, labelled “P09”, on the motherboard as shown in Figure 28.
11. Solder a wire from the “SW” pad on the right trigger button PCB to the test pad, labelled “P08”, on the motherboard as shown in Figure 28.
12. Solder a wire from the “GND” pads on both trigger button PCBs to any ground pad on the motherboard as shown in Figure 28. There are several ground pads, and you can choose any of them.

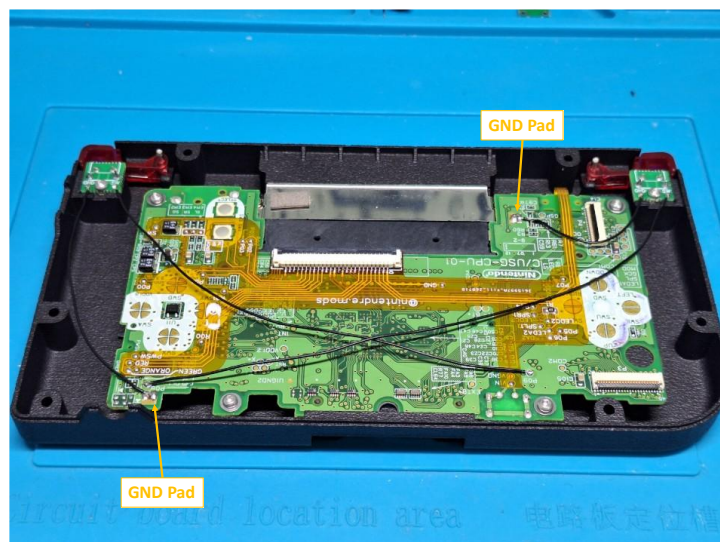


Figure 28: Trigger button PCBs connected to DS Lite motherboard with wires

4.3.3 Final Assembly

1. Connect the LCD screen to the DS Lite motherboard using the 39-contact ribbon cable to connect the FPC connector on the button PCB to the FPC connector for the lower screen on the DS Lite motherboard as shown in Figure 29. For both connectors, the ribbon cable pins should be facing down.
2. Connect the button PCB to the button extender flex PCB using the 26-contact ribbon cable as shown in Figure 29. For both connectors, the ribbon cable pins should be facing down.



Figure 29: Front and rear shell assemblies connected with ribbon cables

3. Close the console by fitting the front shell assembly onto the rear shell assembly as shown in Figure 30.



Figure 30: Front shell assembly and rear shell assembly fitted together

4. Fasten the rear shell using six M2 x 4mm socket head cap screws.
NOTE: If using a resin printed shell, use 2mm x 5mm self-tapping screws instead.
5. Insert the DS Lite battery into the battery bay as shown in Figure 31.



Figure 31: Nintendo DS Lite battery installed in battery bay

6. Close the battery bay using the battery cover.
7. Secure the battery cover using a M2 x 6mm countersunk screw as shown in Figure 32.



Figure 32: Battery cover fixed with countersunk screw

8. Insert a game and turn on the console by pressing the power button (GBA SP brightness button).